Day 1 — Python, SQL & Data Engineering Fundamentals

🎯 Goal

Today I focused on strengthening the fundamentals required for a **Junior Data Engineer – GenAI** role.

The main focus areas were:

- Python programming
- Working with JSON and CSV
- SQL
- REST APIs
- ETL and ELT
- Basic data pipeline concepts

🐍 1. Python for Data Engineering

Topics Covered

- Variables and data types
- Lists and tuples
- Dictionaries
- Loops and conditions
- Functions
- List of dictionaries
- File handling
- Exception handling
- JSON handling
- CSV handling

Important Concepts

Python dictionaries and lists are especially useful for processing API responses and structured data.

Example:

```
student = {
    "name": "Alagu",
    "department": "CSE",
    "mark": 85
}
```

```
print(student["name"])
```

I also practiced working with a list of dictionaries because API responses commonly contain collections of structured records.

📄 2. JSON & CSV

JSON

JSON is a lightweight data-interchange format commonly used by REST APIs.

Python example:

```
import json

data = '{"name": "Alagu", "mark": 85}'

student = json.loads(data)

print(student["name"])
```

CSV

CSV is commonly used for storing tabular data.

I practiced reading CSV files using Python's `csv` module.

🗄️ 3. SQL

Topics Covered

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- COUNT
- AVG
- Primary Key
- Foreign Key
- JOIN

- INNER JOIN
- LEFT JOIN

Example:

```
SELECT department, AVG(mark)
FROM students
GROUP BY department;
```

JOIN Example

```
SELECT s.name, d.department
FROM students s
INNER JOIN departments d
ON s.dept_id = d.dept_id;
```

I focused particularly on **JOIN, GROUP BY and aggregate functions**, as they are important for data-related roles.

🔄 4. ETL

ETL = Extract → Transform → Load

Extract

Collect data from sources such as:

- APIs
- CSV files
- JSON files
- Databases

Transform

Process and clean the data.

Examples:

- Remove duplicates
- Handle missing values
- Change data types
- Filter records
- Perform calculations

Load

Store the processed data in a target system such as a database or warehouse.

🔁 5. ETL vs ELT

ETL

```
Extract
   ↓
Transform
   ↓
Load
```

Transformation happens before loading the data.

ELT

```
Extract
   ↓
Load
   ↓
Transform
```

Data is loaded first and transformed later.

🌐 6. REST API & JSON

I learned the basic structure of REST APIs and how applications exchange data using HTTP requests.

Important HTTP methods:

- GET — retrieve data
- POST — create/send data
- PUT — update data
- DELETE — delete data

Python example:

```python
import requests
```

```
response = requests.get("API_URL")

data = response.json()

print(data)
```

🔗 7. Basic Data Pipeline

I practiced the basic idea of connecting an API with Python processing.

```
REST API
    ↓
JSON Data
    ↓
Python
    ↓
Data Transformation
    ↓
CSV / Database
```

This helped me understand how a simple data pipeline works.

🛠️ Mini Practice Project

API → Transform → CSV Pipeline

The objective was to:

1. Extract data from a REST API
2. Receive the response as JSON
3. Process the data using Python
4. Select the required fields
5. Save the transformed data into a CSV file

This was my first practical exercise related to data engineering.

💡 Key Learnings

Today I understood that Data Engineering is not only about databases.

A simple pipeline can involve:

API → Python → Transformation → Database/File

I also understood the difference between ETL and ELT and how SQL is used to query and transform structured data.

📚 Interview Questions Practiced

- What is ETL?
- What is ELT?
- What is a data pipeline?
- What is a REST API?
- What is JSON?
- What is a primary key?
- What is a foreign key?
- What is JOIN?
- Difference between INNER JOIN and LEFT JOIN?
- Difference between WHERE and HAVING?
- Difference between DELETE, DROP and TRUNCATE?
- How do you read CSV files using Python?
- How do you call an API using Python?

🚀 Next — Day 2

Tomorrow I will focus on **Generative AI and LLM fundamentals**:

- Generative AI
- LLMs
- Tokens
- Context windows
- Prompt Engineering
- LLM APIs
- Temperature
- System/User prompts
- Structured JSON responses

The goal is to move from **Data Engineering fundamentals → GenAI fundamentals**.